

The Factorial vs Sum Example

A short note on poetry, residuals, and why wording matters

Prepared for the VD Newton project

Purpose. This note explains the “factorial vs sum” thought experiment, why it reveals a real layer of interpretation beyond bare formal structure, and how that insight supports the use of residual entries such as E40 in the Newtonian closure triplet.

1. The example

Imagine we are handed a piece of alien code. We can see the symbol string, but we do not yet know how the symbols are interpreted by the alien assembler. There are two candidate readings of the procedure: either it computes the factorial of the input n , or it computes the sum $1 + 2 + \dots + n$.

At first glance, we might think the visible marks should settle the question. But they do not. If we do not already know what the symbols '+' and '*' trigger inside the assembler, then the code is underdetermined from our point of view. The same visible token could be hooked to one operation in one machine and to another operation in a different machine.

Key point. *The code string alone does not fully determine the live procedure for an outside interpreter. The operational role depends on how the symbols are taken.*

2. What this teaches

The lesson is not that logic is useless. Once the interpretation is fixed, logical consequences follow. The lesson is that logic does not always fix its own interface to the world. Some bridge is needed between the visible string and the operation that a competent agent will actually perform.

This bridge is what your example calls poetry. Here 'poetry' does not mean ornament or vagueness. It means the evocative force of wording: which sign, phrase, or image loads the intended rule into a human or machine interpreter. A symbol may be formally acceptable while still failing to trigger the intended procedure in the reader.

3. Link to raw meaning

This fits the older VD draft very naturally. In that draft, a definition is the raw dictionary string, while meaning is the total change in the relevant state caused by the utterance of a word. That makes room for an important distinction: two strings can play nearby formal roles while still producing different updates in the person who reads them.

So a sentence should not be judged only by whether it is technically compatible with the surrounding definitions. It should also be judged by whether it produces the right shift in modelling behaviour. Does it tell the reader what to trust, what to do next, and why the construction matters?

4. Why this supports residual entries

This is where the residual idea becomes important. A residual entry does not necessarily carry the hardest formal criterion in the system. Instead, it can carry the raw meaning that makes the criterion usable. It tells the interpreter what the formal structure is for.

In the closure triplet, E38 gives the hard test: every interaction candidate can be placed in an interaction pair. That is the audit rule. But E40 is doing a different job. It tells the modeller why passing that audit matters: if the system is mechanically closed, then its future behaviour is determined by the equations produced from the interactions internal to it.

Without that residual sentence, the triplet risks sounding like sterile bookkeeping. With it, the intended modelling stance becomes vivid: this is the kind of system for which solving the equations is worth doing, because nothing mechanically relevant has been left out.

5. What I learned from the example

The biggest lesson is that wording can be load-bearing even when it is not the primary formal wall. I initially treated E40 as a softer paraphrase of closure. The factorial-vs-sum example shows why that was too shallow. A reader needs more than a criterion; they need a phrase that loads the use-case of the criterion.

So the poetic layer is real. It is the layer that aligns a definition with the behaviour you want it to evoke. In VD language, it is part of meaning rather than raw definition. That does not weaken rigor. It explains why rigor alone is not enough for successful transmission.

In this case, the residual entry is not excess decoration. It is the sentence that tells the reader: these internally generated equations are not merely present; they are the right things to solve if you want the future behaviour of the system.

6. Working summary

- The alien-code example shows that visible syntax can underdetermine live operation.
- A formal structure still needs an interpretive bridge that loads the intended procedure.
- That bridge is what your example calls poetry: the evocative force of wording.
- Residual entries can carry this raw meaning even when they are not the strict criterion.
- That is why E40 should speak about future behaviour, not merely repeat the bookkeeping condition.

Proposed closing sentence for the Newton entry set:

“A mechanically closed system is a mechanical system whose future behaviour is determined by the equations produced from the interactions internal to it.”